RESEARCH ARTICLE

A Hybrid LECNN Architecture: A Computer-Assisted Early Diagnosis System for Lung Cancer Using CT Images

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Abstract
Lung cancer is one of the most common causes of cancer-related death. Therefore, early diagnosis of this cancer is crucial for planning patient treatment. This paper proposes a hybrid Lung Ensemble Convolutional Neural Network (LECNN) architecture for the computer-aided early diagnosis of lung cancer via CT images. The proposed hybrid approach integrates a transfer learning (TL) mechanism with ensemble learning (EL) on the basis of majority voting. Initially, CNN architectures (GoogLeNet, EfficientNet, DarkNet19, and ResNet18) are trained via TL, and the resulting CNN models are used as inputs in EL. The outputs from all the CNN architectures are evaluated via majority voting to identify the top-performing triple CNN combination, which is then utilized in the hybrid approach. The performance of the proposed method was assessed via the widely used IQ-OTH/NCCD dataset. Additionally, the impact of the elastic transformation method, a data augmentation technique, on performance improvement was investigated in the proposed method. The triple combination of the GoogLeNet, EfficientNet, and DarkNet19 CNN architectures, as part of the EL method in the hybrid approach, achieved superior performance on both the raw and augmented datasets according to the obtained performance results. The performance evaluations revealed that the proposed approach achieved more than a 5% improvement with the augmented dataset compared with the raw IQ-OTH/NCCD dataset, resulting in the highest performance. The proposed hybrid approach achieved 99% accuracy, 98.82% sensitivity, 99.48% specificity, 99.06% precision, and 98.94% F1 score on the augmented IQ-OTH/NCCD dataset. When compared with findings from previous studies using the same dataset, the proposed hybrid approach outperformed state-of-the-art methods. In conclusion, it demonstrates significant potential as a robust tool for computer-aided early lung cancer diagnosis systems and may also contribute to the development of future hybrid approaches in this field.

Keywords Image processing · Deep learning · Convolutional neural network · Ensemble learning · Decision support systems

1 Introduction

The World Health Organization (WHO) maintains that one of the leading causes of cancer-related fatalities globally is lung cancer [1]. The uncontrolled growth of cells in lung tissues, particularly in the airway, leads to this type of cancer. Uncontrolled proliferation results in lung function loss or injury to the surrounding tissue. There are two forms of lung cancer: benign and malignant. A mass that is congenital or acquired does not spread to the tissues and does not represent any threat is referred to as a benign tumor. A malignant tumor is one that damages other tissues and organs due to the unbalanced proliferation of cells in the lung.

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Compared with other cancer types, lung cancer has a higher rate of mortality. Nonetheless, the death rate from lung cancer diminishes with early detection and following adequate treatment planning [1]. For physicians, detecting this cancer early is not easy. Many patients receiving cancer diagnoses at cancer centers do so at advanced stages when there are very few options for treatment [2]. Nevertheless, early diagnosis is made possible by periodic checks of high-risk patients at these institutions. It is best to use a variety of techniques for diagnosis, including imaging and physical examination, for detecting lung cancer. Different types of imaging are extensively employed for the diagnosis of lung cancer. Currently, computed tomography (CT) is utilized for lung scans since it is more sensitive for detecting extremely small nodules in the lung [3]. Fast and accurate analysis of pulmonary nodules via CT images is highly important for computer-aided diagnostic systems in early diagnosis [4].

Interest in medical imaging services, including CT, mammography (MG), ultrasound (USU), and magnetic resonance imaging (MRI), is growing daily as a result of advancements in healthcare technology. The examination of medical images, however, can be challenging and time-consuming when there are not enough professionals in the clinic, considering that images frequently have multiple characteristics [5]. Furthermore, given the extensive workload that experts have, it is probable that the interpretations provided by experts at various points in time may change significantly [6].

Artificial intelligence (AI) is defined as a system that approximates human intelligence and has the capacity to learn from the multitude of data it gathers. Recently, deep learning (DL), an AI technique, has shown notable promise in a variety of clinical applications, including medical procedures used for medical imaging-related early detection, monitoring, diagnosis, and treatment evaluation of various medical conditions [5]. Machine learning (ML) is a subfield of AI that uses data to learn from and then generates predictions on the basis of historical data to carry out a number of complicated tasks. ML employs a variety of learning strategies, including semisupervised, unsupervised, and supervised learning. ML is a foundational learning strategy that attempts to train a machine on how to perform a particular job and deliver correct results by determining patterns. DL, a subfield of ML, is capable of identifying intricate patterns in high-dimensional data by utilizing certain algorithmic structures referred to as neural networks, which are inspired by the organization of the human brain [7]. DL requires no processing and may automatically extract fundamental features from raw inputs without the need for typical ML algorithms [8]. As a result, DL has been favoured in various domains and has outperformed ML techniques in crucial problems, including image recognition [9], speech recognition [10], and prediction of the activity of potential drug molecules [11], and DL has experienced greater success. Owing to its more effective image analysis ability and automatic feature extraction ability, DL has just recently emerged as one of the most commonly used AI techniques for medical image analysis in decision support systems that can be established for early diagnosis in medicine [12]. Imaging techniques have become commonplace in various medical subfields, such as ophthalmology, to diagnose diseases. To achieve this goal, applications for DL-based early diagnosis systems have been constructed to detect an assortment of health issues using medical images acquired via commonly employed imaging methods, such as the detection of retinal diseases [13, 14], breast cancer [15], and brain tumors [16]. In this context, the following section lists and details multiple AI- and DL-based system recommendations for lung cancer detection—the focus of this study—that have been carried out in prior investigations.

In one of the previous studies, Al-Yasriy et al. proposed a computer-aided system utilizing the AlexNet architecture, a type of CNN, as a DL approach for the identification of benign and malignant lung cancer. The IQ-OTH/NCCD [17] dataset was utilized to train and evaluate the AlexNet architecture that is employed in this suggested technique. The suggested method's performance metrics indicated that it had a high accuracy rate of 93.55% in detecting lung cancer [18]. Kareem et al. suggested a computer system that relies on image processing and computer imaging techniques to diagnose lung cancer. The system was based on the IQ-OTH/NCCD lung cancer dataset, which includes CT scans acquired from 110 healthy and tumor-bearing individuals from two Iraqi hospitals. A number of approaches, including the Gaussian filter for image processing, the Gabor filter, and the gray-level co-occurrence matrix, have been implemented in this suggested system to extract features. The ML supervised learning approach (SVM) was then performed to classify all of the acquired features in the classification step. Using a polynomial kernel, the SVM method attained the greatest accuracy of 89.89% on the dataset

[19]. Al-Huseiny et al. presented a TL-based algorithm that incorporates the GoogLeNet architecture as one of the DL techniques for lung cancer diagnosis. The IQ-OTH/NCCD dataset was used to construct and evaluate the recommended approach. They reported that the suggested approach outperformed earlier methods on the IQ-OTH/NCCD dataset [20]. To predict the existence of a lung tumor via CT images, Solyman and Schwenker suggested a method that combines DL, TL, and EL methodologies. In their suggested approach, they used the CNN architectures of VGG-16, ResNet50, InceptionV3, and EfficientNetB7 to develop an EL method based on majority voting. Compared with CNN architectures, better performance was obtained with the devised EL technique [21]. A CNN architecture was demonstrated by Bangare et al. for computer-aided lung cancer classification from CT images via image preprocessing and segmentation techniques [22]. ResNet101 was the CNN architecture that achieved the highest performance in comparison with other models when Huang et al. employed DL models, namely, AlexNet, VGG16, and ResNet101, to make a diagnosis via CT images [23]. Humayun et al. [22] used VGG 16, VGG 19, and Xception CNN architectures using TL method for lung cancer detection and compared their performances by training them on the IQ-OTH/NCCD dataset. According to the performance evaluation, the best performance was obtained with the Xception architecture [42]. In another approach, Nigudgi and Bhyri proposed a hybrid-SVM model based on TL, including CNN models such as AlexNet, VGG and GoogleNet, to detect lung cancer from CT images and reported that the proposed model achieved an accuracy rate of 97% in a performance evaluation [43]. Similar to ref. [43], Venkatraman and Reddy proposed a hybrid model using VGG16 and SVM together to diagnose lung cancer and evaluated its performance [44]. Mostafa et al. proposed an approach for robust detection of lung cancer via the TL approach for feature extraction from CT images followed by SVM for classification. In the performance comparisons, the combination of VGG19 and SVM achieved the highest performance in the proposed approach [45]. In a recent study, Ma et al. presented a new GoogLeNet-AL architecture incorporating adaptive layers for detecting lung cancer, demonstrating its superior performance over traditional CNN models on two public datasets [46]. Yan and Razmjooy developed a new framework for diagnosing lung cancer from CT scans, leveraging the combined application of image preprocessing, the snake optimizer, and CNN algorithms [47]. Gupta et al. introduced a novel method named UDCT, which is based on a modified U-Net architecture, and conducted an ablation study to verify the efficiency and effectiveness of the UDCT framework [48]. Finally, state-of-the-art studies on lung cancer detection have been thoroughly reviewed and discussed. In addition, a comparative performance analysis of the proposed method with previous studies on the basis of performance metrics is provided in the discussion section.

Medical image analysis almost always involves approaches based on a single classifier, as demonstrated above in studies on lung cancer detection. The accuracy performance of the developed method is, however, one of the most crucial challenges associated with an application delivered with AI for medical image analysis. Furthermore, numerous studies on the hybrid or combined use of various ML and DL approaches for enhancing classification performance and speed have been conducted. These methods, which are based on medical image analysis, include AI-based methods for early diagnosis in medicine. For example, the hybrid-SVM method [43] combines an SVM (conventional ML technique) with a CNN architecture to increase the classification accuracy of CT images in early lung cancer detection. In parallel, the explainable FDCNN [24] has been proposed, which combines a CNN architecture (with explainable AI) and an SVM method, demonstrating its superior efficacy. A more recent development in hybrid methodologies is the CNN-based stacking EL method [14], which amalgamates CNNs with the EL technique to advance the classification of medical images. Furthermore, a methodology that combines distinct CNN architectures with the EL method has been devised to detect lung cancer from CT images [21]. In summary, in complex medical image classification tasks, hybrid and combined approaches that use CNNs with ML techniques such as SVM and EL can achieve significantly improved classification performance compared with the use of a single CNN [14, 21, 24, 43–45].

To summarize previous research on lung cancer, various AI-based DL and ML techniques have been developed and presented, obtaining performance metrics in the literature for early detection, usually via CT scans. In this context, motivated by the literature, this study proposes a hybrid LECNN architecture using CNN and EL methods for the automatic detection of lung cancer from CT images. The objective of the proposed method is

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to enhance the performance of individual CNN classifiers by adopting a hybrid approach that integrates CNNs with EL, one of ML method. For this purpose, the EL method, which is based on majority voting, is used in the proposed method and combined with the developed TL method involving CNN architectures. The publicly available IQ-OTH/NCCD [17] lung cancer dataset with CT images, which was recently presented in this study, is used to evaluate the performance of the proposed method and to provide an appropriate performance comparison with methods in the literature. In addition, to address the well-known scarcity problem in medical images and improve performance, a data augmentation technique was applied to the IQ-OTH/NCCD dataset. The performance of the proposed method was also evaluated via both the raw and augmented versions of the dataset. All the results of the experiments carried out on both datasets via the proposed method are presented. Below is a detailed description of the main purposes of this study and its contribution to the literature.

1) To develop a hybrid approach for lung cancer diagnosis via a TL mechanism, including pretrained CNN architectures such as GoogLeNet, EfficientNet, ResNet18, and DarkNet19, alongside an EL method with majority voting,
2) To present a comprehensive performance evaluation of the CNN architectures used on the IQ-OTH/NCCD dataset,
3) Carrying out comprehensive performance tests on the hybrid combination of the CNN architectures used and the EL method,
4) Using elastic transformation, a data augmentation technique for performance enhancement of the proposed method, on the IQ-OTH/NCCD dataset,
5) Proposing of hybrid LECNN architecture as a hybrid approach for lung cancer detection,
6) Providing comparative analyses to showcase the outstanding performance of the hybrid approach, which is based on the combination of the CNN and EL methods.
7) Evaluating the proposed method on IQ-OTH/NCCD dataset, used in the literature, and comparing the findings with the most recent approaches.

Within the remainder of the paper, Sect. 2 supplies an overview of the suggested hybrid approach, the designed mechanism for TL (using pretrained CNN architectures), the EL method, and a descriptive account of the raw and augmented IQ-OTH/NCCD lung cancer datasets. The CNN architectures utilized in the TL process are demonstrated in Sect. 3, together with the results of the tests conducted to determine the efficacy of the suggested approach. The benefits and drawbacks of the study's conclusions are put into perspective with those found in the literature in Sect. 4. Section 2 concludes by reviewing the research findings and outlining future research goals.

2 Materials and Methods

In this study, a hybrid LECNN architecture is proposed for the detection of lung cancer from CT images, and the flow diagram of the proposed method is given in Fig. 1.

2.1 Overview of the Proposed Method

An innovative design for lung cancer detection, designated a hybrid LECNN architecture, was constructed for this study, and the developed architecture was subsequently proposed for the relevant problem in line with performance tests. The flow diagram of the proposed hybrid approach is given in Fig. 1.

The development process of the proposed method comprises five distinct stages, as illustrated in Fig. 1. Initially, the publicly accessible IQ-OTH/NCCD lung cancer dataset, which is frequently cited in the literature, was used to evaluate the performance of the method under study. The first stage involved the application of image preprocessing techniques, such as resizing, to prepare the dataset for the subsequent TL mechanism. In the second

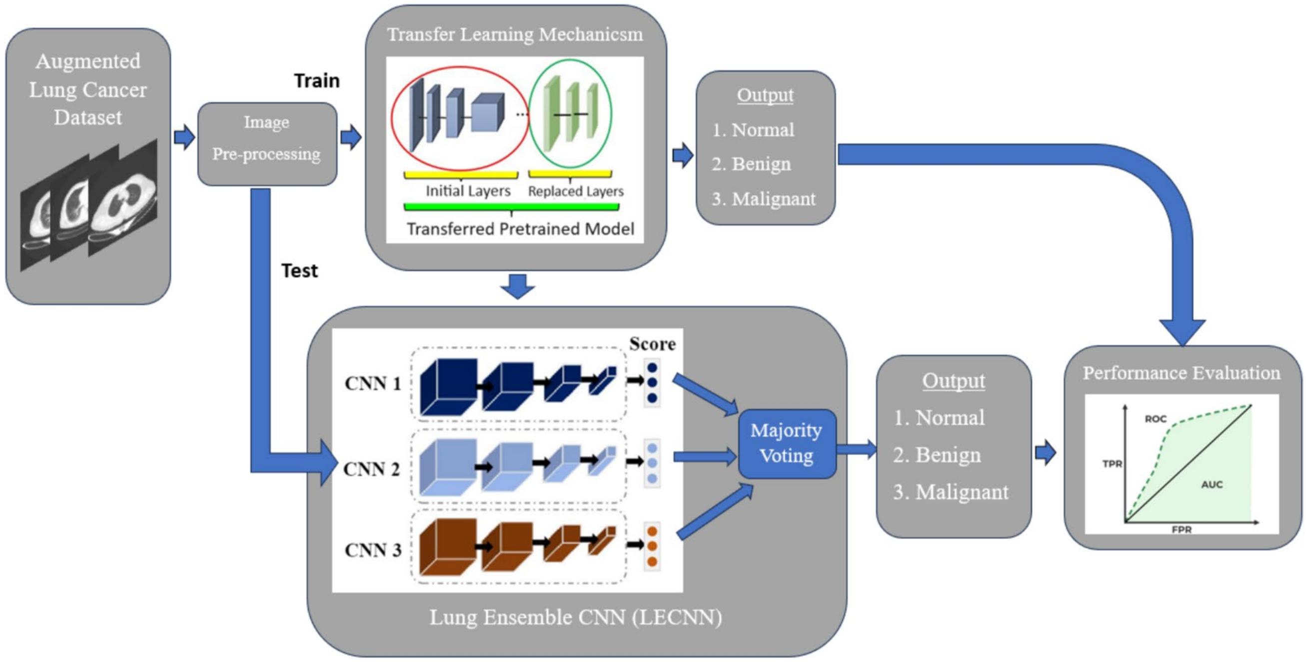

Fig. 1 Flow diagram of the proposed hybrid LECNN architecture

stage, as depicted in Fig. 1, the hyperparameters for the GoogleNet, ResNet18, DarkNet19, and EfficientNet CNN architectures were determined according to the dataset's characteristics, and training was independently conducted for each architecture. Following the completion of training, the performances of the resulting CNN models were evaluated via performance metrics. The trained CNN models were then integrated into the EL method, which constitutes the core of the hybrid approach. In this phase, various combinations of three CNN models were created on the basis of the majority voting principle within the EL method, and the EL method was applied to these combinations. The outcomes of the EL method were assessed via performance metrics, and the findings were systematically documented. Additionally, a further performance comparison was conducted by incorporating a data augmentation technique into the dataset within the context of the proposed hybrid approach. Detailed analyses of the performance impact of data augmentation are provided in the Experiments and Results section. The components and methodologies of the proposed method are elaborated upon in the subsequent subsections.

2.2 Image Preprocessing

To enable the images in the lung cancer dataset to be used as inputs in the CNN architectures deployed in the TL process, as shown in Fig. 1, the images need to be resized to a reasonable size. Because of this, the prescribed dataset was trained on CNN architectures by preprocessing techniques, including image resizing and morphological operations, in the methodology presented in this work. At this stage, the preprocessed dataset obtained with the preprocesses used was made suitable for the training processes carried out in the TL mechanism.

2.3 Transfer Learning Mechanism

When dealing with complicated issues, AI techniques imitate human perception, learning, and reasoning. AI has recently shown impressive results across a wide range of industries. The application of AI techniques in numerous fields has evolved significantly with technological advancements. ML is one of the many sub-branches of AI. A specialized branch of ML, DL, renders the ability to make use of artificial neural network techniques to handle high-dimensional data in multilayered architectures. The multilayered structure and high data processing capacity

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of DL are among its outstanding features, which have allowed it to achieve an outstanding rate of performance in many different scenarios, specifically image classification. For this reason, the best-performing DL techniques are often employed in image classification applications. CNN is one of these DL techniques that has gained popularity since it has outperformed other DL techniques, such as recurrent neural networks, in classification evaluations. Large datasets can be used to address complex problems, and DL can produce outstanding results in these situations. A large dataset or a variety of datasets can be troublesome to acquire in some fields, including medicine. As such, the learned weights of the DL techniques taught for one problem can be applied to another when data are scarce or computational resources are accessible. Using the information learned in one task or one dataset in another task or dataset is known as TL [25].

TL leverages learned patterns acquired from several problems to address a computer imaging problem. Moreover, it can offer models for problem solving in this paradigm that demonstrate greater success and learn more quickly with less training data [26]. TL performs very effectively in determining new tasks and has recently achieved significant success in problems with datasets that do not have sufficient diversity, such as disease diagnosis in medicine [13, 27, 28].

High-capacity CNNs are the foundation of many pretrained models that are favoured for TL applications in DL [29]. It has been confirmed that CNNs perform better overall in computer vision-based tasks [30]. The capacity of CNNs for excellent performance and their simplicity of training are two of the key elements that contribute to their growing popularity. The ImageNet large dataset was prepared for the Large-Scale Visual Recognition Competition (ILSVRC), which was held periodically starting in 2012, and pretrained CNN architectures such as VGGNet and AlexNet have proven their success on this dataset in competitions. In recent years, these designs have yielded encouraging outcomes when employed for medical image analysis via TL, according to comprehensive analyses [31]. Accordingly, the parameters of well-trained CNN models on ImageNet data (e.g., GoogleNet [32] and ResNet [33]) can be transferred to a targeted CNN model to solve a medical imaging classification problem. Using pretrained CNN models, a TL mechanism was established in this study for the diagnosis of lung cancer, as illustrated in Fig. 1. Table 1 also provides a comparison of the fundamental characteristics of the CNN architectures utilized in this TL process.

Four distinct pretrained CNN architectures were favoured in the TL mechanism as part of the methodology presented in the present article, as indicated in Table 1, which may result in exceptional performance in medical image analysis.

In 2014, during the ILSVRC event for ImageNet, Google researchers unveiled GoogLeNet, a 22-layer deep CNN architecture that is a variation of the startup network. Owing to the initial modules introduced with this architecture, features of the network at different scales were achieved, and performance improvement was achieved without an increase in computational requirements [32]. The residual neural network (ResNet) was introduced by He et al. and was developed to facilitate the training processes of very deep neural networks [33]. ResNet has an edge over other DL models because it incorporates residual blocks with residual functions. Additionally, the network may learn the residuals, or the difference between an input and output of a specific layer, owing to the skip connections that are offered by ResNet. Residual learning serves this function by improving the effectiveness of training deep neural networks, such as ResNet. YOLOv2's backbone network, DarkNet19, is a CNN design with 19 convolution layers. Unlike other CNN architectures, this one uses the global average pooling layer for prediction and operates in network-in-network logic [34]. EfficientNet is a CNN design that, in contrast

Table 1 Basic features of the structures of the pretrained CNN architectures

Network	Depth	Size (MB)	Parameters (Millions)	Image input size
GoogleNet [32]	22	27	7.0	224 × 224
ResNet18 [33]	18	44	11.7	224 × 224
DarkNet19 [34]	19	78	20.8	256 × 256
EfficientNet [35]	82	20	5.3	224 × 224

to conventional implementations, applies a composite coefficient to calculate the depth, resolution, and width dimensions equally. Additionally, it has MobileNetV2's inverted bottleneck residual blocks at its core. With the use of EfficientNet TL, it achieved state-of-the-art accuracy in image classification problems by providing good transfer with fewer parameters [35].

The performances of the four CNN architectures are evaluated on the basis of the lung cancer dataset used in the current study, which results from the use of pretrained CNN architectures, the specifics and explanations of which are reported in Table 1 above. By evaluating the performance of the proposed hybrid LECNN architecture, a variety of triple CNN combinations were created for the proposed method via these four trained CNN architectures. The combination of the three CNN architectures that achieved the highest performance was then used in the structure of the proposed hybrid approach.

2.4 Ensemble Learning

An ML technique identified as EL combines several models to generate a more effective model [36]. The basic principle of EL is to ensure that the overall performance is improved compared with that of individual models by using the predictions of more than one model. EL is generally preferred to improve the accuracy, stability and generalizability of ML models [14]. EL methods are based on bagging, boosting and majority voting. However, the voting-based EL method is an effective method used to increase the overall accuracy by combining a set of classifiers [37]. Equation (1) provides a mathematical representation of the majority voting-based EL technique.

$$y = \text{mode}\left\{C_1(x), C_2(x), \ldots ..C_m(x)\right\}; y = \text{mode}\{0,0,1\} = 0 \tag{1}$$

Here, in this case, an input is denoted by x, a classifier by C, the total number of classifiers by m, and the final classification label by y. In this equation, the prediction of each C classifier is determined, and then the final class label is predicted through majority voting. As noted above, three sample classifiers were integrated into a majority EL model, and classification labels were assigned; as a result, the final class was determined to be 0 by majority. In the current study, four CNN architectures trained for the detection of lung cancer were combined to form triple CNN combinations. The created triple CNN combinations were tested separately on the EL method on the basis of a majority vote. Finally, the triple CNN combination with the highest performance was used in the proposed LECNN architecture.

2.5 Proposed Hybrid LECNN Architecture

As shown in Fig. 1, a hybrid LECNN architecture was designed for the diagnosis of lung cancer in this study. Two key components comprise the fundamental structure of the created hybrid method: the implementation of the majority-based EL method and the construction of the TL mechanism. In this study, the performance of four CNN architectures in total was tested, and then the prediction outputs obtained from three CNN architectures were used using the CNN and EL together in the proposed method. Given that there is the possibility of generating an equal number of prediction outputs when the outputs from four CNN designs are merged together, this might result in system instability. Consequently, to prevent instability in the suggested method and to make distinct performance comparisons, triple CNN combinations were constructed using four distinct CNN architectures trained on the TL mechanism. Later, the prediction outputs of the created triple CNN combinations obtained from the test section of the lung cancer dataset were used as inputs in the majority-based EL method. A performance evaluation was performed according to the outputs obtained after applying the EL method to these triple CNN combinations. The triple CNN combination that functioned most successfully was identified following the performance evaluation, and the chosen CNN combination

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was then incorporated into the hybrid suggested method's final architecture. The classification performance of the proposed hybrid approach is tested through extensive experiments on raw and augmented versions of the commonly used IQ-OTH/NCCD lung cancer dataset, and its remarkable performance is compared with that in the literature.

On the basis of extensive performance evaluations, the proposed method aims to obtain a hybrid model with stable and improved performance by combining the outputs of CNN architectures with those of the EL method for lung cancer detection. In addition, the proposed hybrid approach shows that a more effective combined model can be obtained with the hybrid use of the CNN and EL approaches rather than approaches using a single classifier that achieves low accuracy, as seen in the literature. With the proposed hybrid LECNN approach, comprehensive performance evaluations are performed on the IQ-OTH/NCCD dataset, and performance metrics are presented that demonstrate its robust performance compared with the literature.

2.6 IQ-OTH/NCCD Lung Cancer Dataset (Raw/Augmented)

The lung cancer dataset from the Iraq-Oncology Teaching Hospital/National Center for Cancer Diseases (IQ-OTH/NCCD) was employed for all the experiments conducted for this investigation [18]. This dataset, which includes CT images of patients who are healthy or who have been diagnosed with lung cancer at different stages, was gathered over the course of three months in 2019 from many hospitals in Iraq that specialize in lung cancer. This dataset contains a total of 1097 CT images with 512×512 resolution and jpeg format, representing CT scan slices of 110 cases varying according to age, sex, educational status, place of residence and living conditions, and the labelling of this dataset was carried out by expert oncologists and radiologists. The CT images in this dataset were classified in three different ways: benign (15 cases), malignant (40 cases) and normal (55 cases). Furthermore, for further research on lung cancer detection, the authors have made this dataset freely available in the Kaggle repository [17]. It is evident from the raw IQ-OTH/NCCD dataset's details that CT images are not split into two categories for the sake of training and testing AI-based techniques. Therefore, 80% of this dataset was randomly assigned as training and the remaining 20% as testing, and the same splitting ratio was preferred in all the experiments performed in this study. This procedure was carried out to standardize the training and performance evaluation processes of all methods developed in this study.

Currently, the collection of large datasets for medical image analysis is still a challenge due to privacy concerns and labelling costs. In this context, various data augmentation techniques based on the transformation of original data and the generation of artificial data to greatly increase the amount and diversity of available data without collecting new samples are used to solve this problem [39]. Furthermore, these methods are employed primarily to increase the diversity of the dataset and mitigate the problem of overfitting, especially in the context of deep learning methods. Recently, the use of data augmentation techniques in various image classification tasks, such as classification and segmentation, has positively impacted the performance of trained AI models [39]. Therefore, data augmentations such as random rotation, scaling, elastic transformation, rotation, and flipping have been utilized in the most advanced studies on CT images [40]. As a popular technique, elastic transformation is employed to extend the dataset, thereby improving the robustness and accuracy of AI models. In this context, an augmented version of the IQ-OTH/NCCD dataset with more than 10 times more CT images than the original dataset, which was recently obtained via elastic transformation, has been made publicly available [41]. In a different context, the augmented version of the IQ-OTH/NCCD dataset, unlike the original dataset, is organized into two sections: training and testing. In conclusion, the performance of the proposed hybrid approach was assessed on both the raw and augmented versions of the IQ-OTH/NCCD dataset within the framework of small and large datasets, and the results were reported. Finally, example images from each class contained in the raw and augmented IQ-OTH/NCCD datasets are shown in Fig. 2. In addition, the detailed data distributions by class for these datasets are given in Table 2.

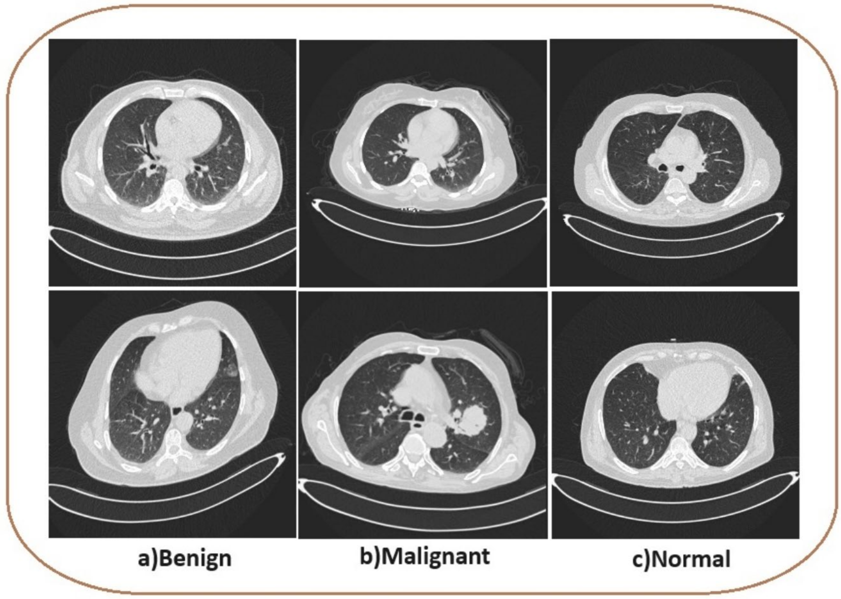

Fig. 2 Sample images for each class from the IQ-OTH/NCCD lung cancer datasets: **a** benign, **b** malignant, and **c** normal (the first row is the raw version of this dataset, and the second row is the augmented version)

Table 2 Details of the raw and augmented IQ-OTH/NCCD lung cancer datasets

IQ-OTH/NCCD lung cancer dataset				
Raw		Augmented		
Class	Image	Class	Train	Test
Normal	416	Normal	3.264	1312
Benign	120	Benign	2.158	962
Malignant	561	Malignant	3.080	1408
Total	1.097	Total	8.502	3.682

3 Experiments and Results

A hybrid LECNN architecture for cancer diagnosis was developed in this study. The first phase in developing this established strategy involves the use of the TL mechanism. Some pretrained CNN architectures used in the TL mechanism were retrained on the IQ-OTH/NCCD lung cancer dataset, and then various triple CNN combinations were created from the trained CNN architectures. The created triple CNN combinations were used as inputs to the EL method, as shown in Fig. 1, and after the performance tests, the CNN combination with the highest performance was used in the structure of the proposed hybrid LECNN architecture. In addition, the effects of the augmented dataset obtained via the data augmentation technique compared with the raw dataset used in this study on performance improvement are investigated, and the findings obtained are presented. The subheadings in this chapter include an explanation of all the information related to the performance measures, experimental conditions and experiments carried out in this study to validate the proposed method.

3.1 Experimental Setting

All experiments in the development process of the hybrid approach in this study were carried out on the MATLAB 2023a software environment on a computer with an Intel (R) Core i7 & 2.6 GHz processor running Windows 10 with an NVIDIA GTX 1650Ti graphics card.

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3.2 Hyperparameter Selection

Hyperparameter selection is one of the most critical processes within DL and has the greatest impact on CNN architecture performance and training procedures (13). For CNN training, it is mandatory to determine several imperative hyperparameters, including the loss criterion, the optimization method, the learning rate, L2 regularization, the batch size, the gradient decay factor and the number of epochs. In this study, hyperparameters that stand out according to extensive performance evaluation findings in studies conducted for CNN-based medical image analysis were preferred [13]. The specifics of these established hyperparameters are provided in Table 3. The same hyperparameters were established for each experiment to enable a comparison of the CNN architectures' performances in appropriate environments.

On the basis of the hyperparameters listed in Table 3, the CNNs utilized in this investigation were prepared for training. Multiple layers, including the input layer, pool layer, fully connected layer, convolution layer, and output layer, make up CNNs [7]. The CNN's convolution layers apply filters to the input images to extract features during training. To reduce the computational cost, feature reduction is provided by the pooling layers that are used later. The acquired features are subsequently gathered in the fully connected layer, and ultimately, the output layer executes the ultimate prediction. Once the training hyperparameters have been established, the CNN architecture's training procedure uses the weights acquired via gradient descent and backpropagation to determine which problem-specific filters are the most acceptable. In the final stage of training the CNN, one of the frequently used hyperparameters in the classification layer is the loss criterion. The loss criterion indicates the deviation between the predicted output of the network and the actual data labels, and this criterion works together with the Softmax function. In this study, cross-entropy loss was preferred, and the mathematical expression of the loss criterion used is given in Eq. (2).

$$H(p,q) = -\sum_{x} p(x)\log q(x) \tag{2}$$

In this case, p is the output for the categorical class, q is the Softmax output, and x is the number of classes. This formula yields a final output, p, which is a summary of all the projected probabilities in the CNN. Many nonlinear layers, such as ReLu, are employed in CNNs and DL for feature extraction and modification. Thus, to improve the efficacy of weight updates during training, the Adam [38] optimization technique, which is based on stochastic gradient descent (SGD), was employed in this study. The Adam optimizer achieves good performance in various training circumstances by adaptively modifying the learning rate to compensate for sensitivity to gradients. In this context, interest in the use of the Adam optimizer has recently increased because of its superior performance in medical image analysis problems.

Table 3 Details of the hyperparameters used for training the CNN architectures

Parameter	Value/type
Loss criterion	Crossentropyex
Optimizer	Adam
Learning rate	0.0001
L2 regularization	0.0001
Batch size	64
Gradient decay factor	0.9
Epoch	50
Output	Normal, benign and malignant

3.3 Performance Metrics

The study adopted a confusion matrix, a metric that is frequently employed for medical image classification problems, to assess the efficacy of the techniques used. There are four fundamental terms in the confusion matrix, namely, true positive (TP), true negative (TN), false positive (FP) and false negative (FN), and the accuracy, specificity, precision (or true positive rate), precision, false-positive rate (FPR) and F1 score metrics are calculated through these terms. These metrics are used to evaluate the performance of a classification model in more detail. The mathematical expressions for the calculation of these metrics are given in Eqs. (3)–(8).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{4}$$

$$\text{Sensitivity (TPR)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

$$\text{F}_1\text{score} = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{7}$$

$$\text{False Positive Rate(FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{8}$$

The receiver operating characteristic (ROC) curve is one of those variables considered to determine how well the AI classification model performs. The area under the ROC curve, or area under the curve (AUC), represents how well a binary classifier can distinguish between classes. AUC values range from 0 to 1, and when this value approaches 1, it is an indication that an AI model is achieving effective performance [26]. In summary, an ROC curve shows the true positive rate (TPR) versus the false-positive rate (FPR) at different classification thresholds.

3.4 Performance Evaluation of Pretrained CNN Architectures with the TL Mechanism

The suggested hybrid strategy included the development of a TL mechanism for lung cancer detection, as is apparent from the study's flow diagram. The IQ-OTH/NCCD lung dataset was initially identified to train the CNN architectures utilized in the TL mechanism. The learned weights of the CNN architectures for GoogleNet, ResNet18, DarkNet19, and EfficientNet were transferred via the TL approach in this phase after this dataset. The specifics of these CNN designs are shown in Table 1 and are accessible in the MATLAB environment. As shown in Table 1, the dataset needs to be prepared so that CNN architectures can use it as input before training begins. To achieve this goal, CT images in the IQ-OTH/NCCD dataset were resized to 224×224 for the GoogLeNet, ResNet18 and EfficientNet architectures and to 256×256 for DarkNet19 in the preprocessing step and were made suitable for training CNN architectures. To prepare for the training of the CNN architectures used in the TL mechanism, in the final process, the hyperparameters in Table 3 were

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Table 4 Performance metrics of the GoogLeNet, DarkNet, ResNet18 and EfficientNet architectures on the raw IQ-OTH/NCCD dataset (The best values are shown in bold style)

Method	Accuracy	Sensitivity	Specificity	Precision	F1 score	AUC
GoogleNet	0.8995	0.8623	0.9587	0.8191	0.8302	0.9679
DarkNet19	0.9269	0.9160	0.9712	0.8610	0.8765	0.9687
ResNet18	**0.9315**	**0.9398**	**0.9744**	**0.8718**	**0.8875**	**0.9846**
EfficientNet	0.9087	0.8901	0.8901	0.8364	0.8364	0.9781

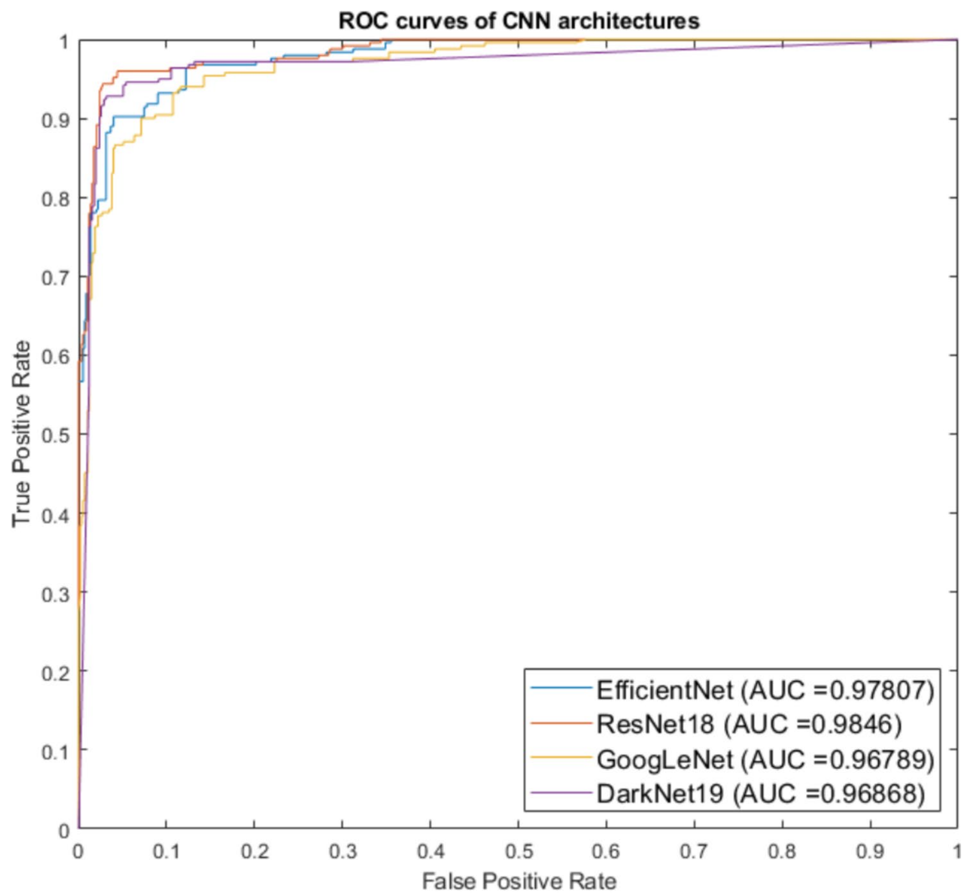

Fig. 3 ROC curves of the CNN architectures obtained on the raw IQ-OTH/NCCD dataset

adjusted, and then the training of the CNN architectures started. After all training was completed, the CNN architectures used were tested on the IQ-OTH/NCCD dataset reserved for testing in this study. As a result, as a result of the test, a confusion matrix was created for each CNN architecture, and a comparison of the performance metrics calculated from the created confusion matrices is given in Table 4. Furthermore, Fig. 3 provides a performance comparison of the trained CNN architectures on the ROC curve of the AUC metrics listed in Table 4.

Another performance evaluation with the TL mechanism was conducted via an augmented version of the IQ-OTH/NCCD dataset. The augmented dataset was given as input to the TL mechanism, and the CNN architectures were trained separately according to the hyperparameters specified in Table 3. The augmented dataset was given as input to the TL mechanism, and the CNN architectures were trained separately according to the hyperparameters specified in Table 3. After the training ends, confusion matrices were created, and

Table 5 Performance metrics of the GoogLeNet, DarkNet, ResNet18 and EfficientNet architectures on the augmented IQ-OTH/NCCD dataset (The best values are shown in bold style)

Method	Accuracy	Sensitivity	Specificity	Precision	F1 score	AUC
GoogleNet	97.69	0.9733	0.9883	0.9769	0.9749	0.9906
DarkNet19	98.59	0.9829	0.9927	0.9871	0.9848	0.9927
ResNet18	97.64	0.9731	0.9880	0.9764	0.9746	**0.9984**
EfficientNet	**98.75**	**0.9845**	**0.9936**	**0.9883**	**0.9862**	0.9966

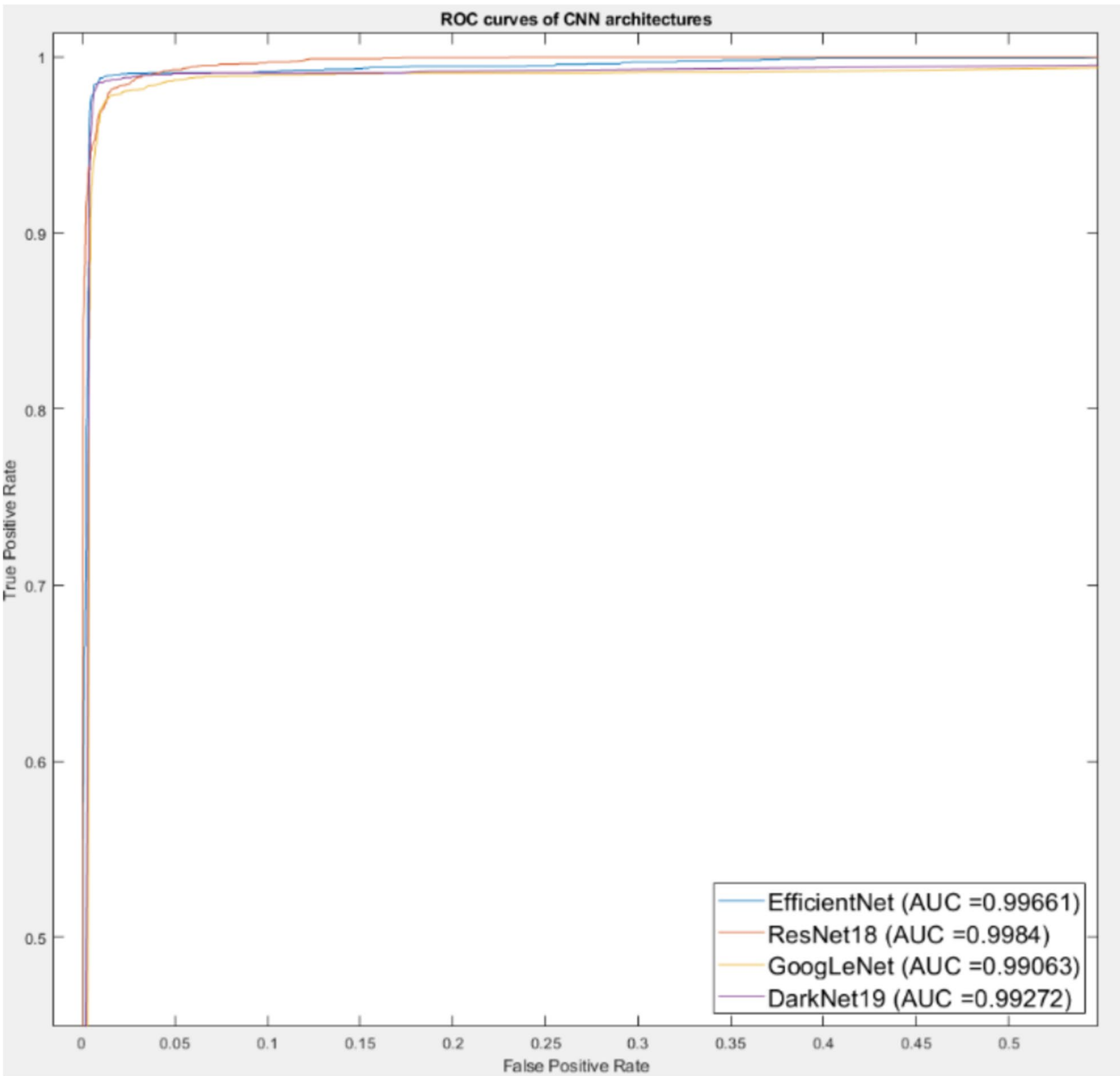

Fig. 4 ROC curves of the CNN architectures obtained on the augmented IQ-OTH/NCCD dataset

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Table 6 Performance comparison of the proposed hybrid approach on the raw IQ-OTH/NCCD dataset (The best values are shown in bold style)

Netw. 1	Netw. 2	Netw. 3	Accuracy
DarkNet19	ResNet18	EfficientNet	94.52
DarkNet19	ResNet18	GoogLeNet	94.52
ResNet18	EfficientNet	GoogLeNet	94.06
GoogLeNet	**EfficientNet**	**DarkNet19**	**94.98**

Table 7 Performance comparison of the proposed hybrid approach on the augmented IQ-OTH/NCCD dataset (The best values are shown in bold style)

Netw. 1	Netw. 2	Netw. 3	Accuracy
DarkNet19	ResNet18	EfficientNet	98.88
DarkNet19	ResNet18	GoogLeNet	98.56
ResNet18	EfficientNet	GoogLeNet	98.67
GoogLeNet	**EfficientNet**	**DarkNet19**	**99.00**

performance measurements were performed. The performance metrics of the trained CNN architectures on the augmented dataset are given in Table 5, and their ROC curves are also shown in Fig. 4.

In Table 4, the performance metrics of the CNN architectures trained on the IQ-OTH/NCCD dataset are presented, and as a result of the comparison, the ResNet18 architecture was one of the methods that showed the best performance in all presented performance metrics compared with the others. Following the ResNet18 architecture, DarkNet19 achieved the second-best performance. The AUC value of ResNet18 is close to 1; accordingly, ResNet18 is capable of accurately differentiating between the benign, malignant, and normal classifications that were employed in this investigation to detect lung cancer.

According to Table 5, EfficientNet achieved superior performance in all metrics except AUC, whereas ResNet achieved the highest performance in the AUC metric, with a value close to 1. Additionally, as shown in Fig. 4, all trained CNN architectures demonstrated effective performance, showing potential as the most robust classifiers with AUC values close to 1. However, according to Tables 4 and 5, ResNet18 has proven to be one of the methods that achieves the highest performance on both the raw and augmented datasets. The performance evaluation indicated that, compared with the raw data, the augmented dataset provided a performance improvement exceeding 5% in terms of the highest accuracy. This demonstrates that the data augmentation technique is a robust method for enhancing performance in medical imaging problems, particularly in lung cancer detection.

In this study, the trained CNN architectures are used as inputs to the EL method as part of the hybrid approach. In accordance with Fig. 1, three CNN architectures are taken as inputs for the EL method. In this regard, various combinations of three out of four trained CNN architectures were created and used as inputs for the EL method on the basis of majority voting. In the next step, the performance of these CNN combinations used in the EL method was evaluated.

3.5 Performance Evaluation of the Proposed hybrid LECNN Architecture

The performance of the proposed hybrid approach was evaluated on both the raw and augmented IQ-OTH/NCCD datasets. In the proposed hybrid approach, the majority-based EL method was applied to the test dataset to triple CNN combinations created from four different CNN architectures used in the TL mechanism. First, the majority-based EL method as the basis of the hybrid approach was applied to triple CNN combinations formed by four different CNN architectures, which were trained on the raw dataset and then tested on the test part of the dataset. After the test, the accuracy performance metric of each CNN combination was calculated, and the accuracy performances obtained are presented in Table 6.

Table 6 highlights that the GoogLeNet, EfficientNet, and DarkNet19 architectures from the combinations of triple CNN networks produced the best results in the suggested strategy. The performance evaluation of the proposed hybrid approach is also performed on the augmented dataset. For this purpose, triple combinations created from CNNs trained with the augmented dataset were used as inputs for the EL method, as shown in Table 6. In the next step, the EL method was applied to these CNN combinations, which were then tested on the test portion of the dataset. According to the test results, the accuracy performances obtained from each CNN combination are given in Table 7.

According to Table 7, similar to Table 6, the highest performance was achieved by the combination consisting of the GoogLeNet, EfficientNet, and DarkNet19 architectures. On the basis of Tables 6 and 7, confusion matrices were created to analyse in more detail the performance parameters of the CNN combination that achieved the highest and most effective performance on both the raw and augmented datasets. The created matrices are shown in Fig. 5. All the performance metrics calculated via the confusion matrix of the proposed hybrid approach are presented in the performance comparison table in the discussion section.

As shown in Fig. 5, the proposed method demonstrates a minimal number of misclassifications in classes such as benign and malignant on both raw and augmented datasets, while achieving flawless classification performance in the malignant class. Based on the experiments conducted, the proposed hybrid approach, which integrates the EL method with a triple CNN combination of trained GoogLeNet, EfficientNet, and DarkNet19 architectures, achieved remarkable results on both the raw and augmented IQ-OTH/NCCD datasets. The performance evaluation using the augmented dataset revealed an improvement exceeding 4% compared to the raw dataset. These findings underscore that data augmentation via elastic transformation can significantly enhance lung cancer detection performance. Furthermore, the superior performance achieved through the combined use of GoogLeNet, EfficientNet, and DarkNet19 architectures on both datasets demonstrates the effectiveness of these three architectures in lung cancer diagnosis. Finally, as evidenced by Tables 6 and 7, this study highlights that the hybrid use of CNNs and ML methods, such as EL, can achieve more effective performance compared to relying on a single classifier.

4 Discussion

In this study, on the basis of the detailed literature review given in the introduction, a performance comparison of the proposed hybrid method with state-of-the-art methods on the IQ-OTH/NCCD lung cancer dataset is given in Table 8.

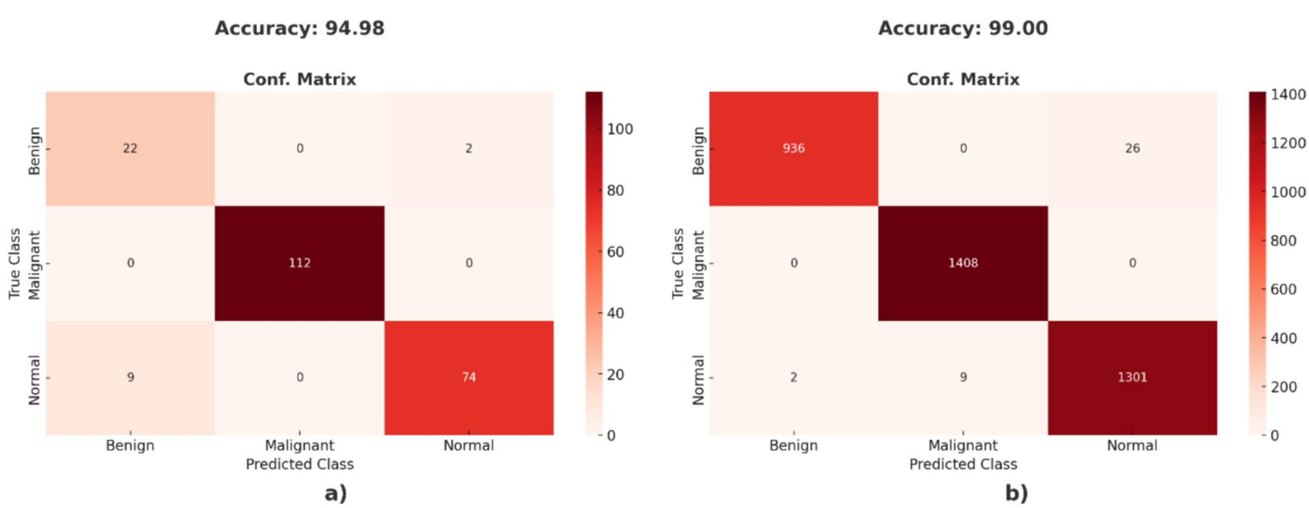

Fig. 5 Confusion matrices of the proposed hybrid approach obtained according to different versions of the IQ-OTH/NCCD dataset: **a** raw **b** augmented

Table 8 Performance comparison of the proposed hybrid approach with the state-of-the-art methods on the IQ-OTH/NCCD dataset (The best values are shown in bold style)

Source	Method	Accuracy	Sensitivity	Specificity	Precision	F1 score
Al-Yasriy et al. [18]	AlexNet	93.55	95.71	95	97.10	96.40
Kareem et al. [19]	SVM	89.89	97.14	97.5	98.55	97.84
AL-Huseiny and Sajit [20]	GoogLeNet	94.38	95.08	93.7	–	–
Solyman and Schwenker [21]	EL	92.80	–	–	–	–
Bangare et al. [22]	CNN	86.42	86.11	86.72	–	–
Huang et al. [23]	ResNet101	88.69	–	–	–	–
Humayun et al. [42]	Xception	89.68	89.68	–	89.68	89.68
Nigudgi and Bhyri [43]	hybrid-SVM	97	93.33	98.33	96.66	–
Venkatraman and Reddy [44]	VGG16 + SVM	89.36	91.78	–	90.1	–
Mostafa et al. [45]	VGG19 + SVM	98	–	–	–	–
Ma et al. [46]	GoogLeNet-AL	97.32	98.20	97.63	**99.45**	98.82
Yan and Razmjooy [47]	CNN + Snake Optimizer	96.58	95.38	94.08	84.16	91.53
Gupta et al. [48]	UDCT	96.82	97.50	98.40	98.70	98.24
Proposed method	**LECNN**	**99.00**	**98.82**	**99.48**	99.06	**98.94**

Table 8 shows that the proposed hybrid approach outperforms the state-of-the-art methods in all performance metrics except the precision metric. In terms of the precision metric, the GoogLeNet-AL [46] method, which is a similar approach to our proposed method, achieves superior performance. Compared with hybrid methods [43–48], which have been recently validated with the IQ-OTH/NCCD dataset for lung cancer detection, the proposed hybrid approach achieved the most superior and overwhelming performance.

The proposed hybrid approach illustrates that employing multiple classifiers in combination, rather than a single classifier as commonly utilized in the literature, results in a robust classification mechanism. Additionally, performance enhancement was examined by testing both the raw and augmented versions of the dataset with the proposed method. Summarizing the experiments and performance comparisons, the hybrid approach—primarily involving the combined use of CNN and EL—demonstrated an approximately 2% performance increase with the raw dataset and a 0.25% increase with the augmented dataset compared with single CNN classifiers. Furthermore, the performance evaluation of the proposed hybrid approach with data augmentation techniques revealed a significant performance improvement exceeding 5% with the augmented dataset relative to the raw dataset. This finding highlights the indispensable impact of the elastic transformation technique in data augmentation on the performance of DL-supported hybrid approaches.

In summary, the comprehensive experiments conducted for the proposed hybrid approach and the results obtained confirm its impressive and robust performance. Additionally, as evidenced by Table 8, the performance comparison with state-of-the-art studies using the same dataset further supports this claim, highlighting the superior performance of the proposed hybrid approach. In this context, the proposed hybrid approach has been included in the literature as a state-of-the-art method and a powerful tool for decision support systems that can be developed early for lung cancer detection.

5 Conclusion

This study presents a hybrid LECNN architecture as an advanced framework for developing computer-aided diagnostic systems aimed at the early detection of lung cancer. As lung cancer remains one of the most lethal malignancies worldwide, early diagnosis plays a pivotal role in significantly improving clinical outcomes, reducing healthcare costs, and enhancing overall public health. The proposed LECNN architecture integrates CNNs trained via TL mechanism with EL technique, establishing a robust and reliable classification methodology.

The performance of the LECNN architecture was comprehensively evaluated using the IQ-OTH/NCCD dataset, demonstrating its superior capabilities in lung cancer detection. The hybrid approach, which utilized state-of-the-art CNN architectures such as GoogLeNet, EfficientNet, and DarkNet19, consistently achieved superior results across both raw and augmented datasets. The application of elastic transformation as a data augmentation technique proved instrumental, yielding a remarkable accuracy of 99% and an improvement exceeding 5% over the raw dataset. These findings highlight the pivotal role of data augmentation in improving the effectiveness of medical image classification frameworks, particularly in addressing the complexities inherent to medical imaging. Furthermore, the LECNN architecture consistently outperformed individual CNN classifiers and demonstrated statistically significant improvements when compared to prior studies utilizing the IQ-OTH/NCCD dataset. The architecture was validated against multiple key performance metrics, including accuracy, sensitivity, specificity, precision, and F1 score, solidifying its standing as a robust and effective diagnostic tool. These findings highlight the potential of the LECNN framework to address the challenges associated with early detection in lung cancer, thereby contributing to advancements in computer-aided diagnostic technologies.

In conclusion, the hybrid LECNN architecture represents a significant contribution to the field of medical image analysis, offering a state-of-the-art approach to lung cancer detection. By combining CNNs with EL method, the LECNN architecture paves the way for future innovations in medical imaging and diagnostic systems. Future research endeavors should focus on evaluating the generalizability of the proposed architecture across diverse medical datasets and exploring its real-world applicability in clinical settings, ensuring its broader adoption in healthcare practices.

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Author Contributions IK: methodology, conceptualization, software, validation, writing—original draft, writing—review and editing. GEG: methodology, conceptualization, supervision, validation, writing—review and editing. All the authors have read and agreed to the final version of the manuscript.

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Data Availability The raw and augmented IQ-OTH/NCCD datasets used in this study are publicly accessible (available at https://www.kaggle.com/datasets/hamdallak/the-iqothnccd-lung-cancer-dataset and https://www.kaggle.com/datasets/aleksandarcvetanov/iq-othnccd-lung-cancer-augmented-dataset).

Declarations

Conflict of interest The authors declare no competing interests.

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